Cover letter

Dear Professor Haueisen,

Please consider our Research Article, “Matched-label analysis identifies spatial refitting and alignment gains in motor-imagery EEG,” for Biomedical Engineering / Biomedizinische Technik.

The study quantifies the additional value of spatial refitting and support-only alignment when a new recording session provides approximately 20 labels. In the predefined 12-participant cohort, full common-spatial-pattern/linear-discriminant-analysis refitting added 5.47 percentage points of balanced accuracy beyond classifier-only recalibration (95 % participant-bootstrap interval 2.17–8.91). It accounted for 73.7 % of the total mean improvement over frozen decoding. Trial-level prediction transitions and longitudinal participant analyses explain this component contribution.

In the secondary comparison in nine BNCI2014-001 participants, support-only EA-CSP improved on full refitting by 5.65 points (3.15–8.70; Holm-adjusted p=0.01172), with positive differences in all nine participants. Full refitting and classifier-only recalibration differed by +0.19 points (95 % interval −1.85 to 2.13).

Exploratory experiments in the six development participants extended the comparison to neural normalization. At the 20-label cap, weight adaptation with fixed source normalization achieved 1.50–1.91 percentage points higher balanced accuracy than adaptive normalization across two source-model regimes. The complete budget profiles show how this effect varies with calibration size.

The manuscript fits the journal’s biosignal-processing and neuro-engineering scope. Public data provenance, the regulatory basis for secondary use, code and data availability, and substantive AI assistance are documented in the separate ethical and legal declarations.

Both authors have approved this manuscript and its submission. The manuscript is not under consideration by another journal. Five suggested reviewers are provided in the accompanying submission information.

Sincerely,

Ziyuan Zhu

Corresponding author, on behalf of Ziyuan Zhu and Sihao Liu

College of Artificial Intelligence Medicine, Chongqing Medical University, Chongqing, China

zzy2630816871@gmail.com

ORCID: https://orcid.org/0009-0005-1153-4926